

LEARNING MANAGEMENT SYSTEM FOR RURAL SCHOOL

**A Project Report submitted in partial fulfillment of the requirements for the
award of the degree of**

BACHELOR OF TECHNOLOGY

IN

COMPUTER SCIENCE AND ENGINEERING

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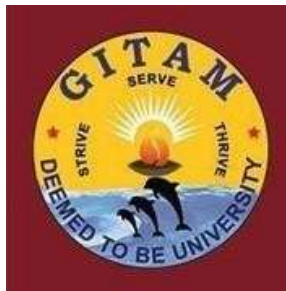
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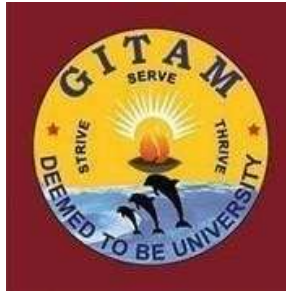
**DEPARTMENT OF COMPUTER SCIENCE & ENGINEERING GITAM
(Deemed to be University)**

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DECLARATION

I/We, hereby declare that the project report entitled “ **Learning Management System for Rural school**” is an original work done in the Department of Computer Science and Engineering, GITAM School of Technology, GITAM (Deemed to be University) submitted in partial fulfillment of the requirements for the award of the degree of B.Tech. in Computer Science and Engineering. The work has not been submitted to any other college or University for the award of any degree or diploma.

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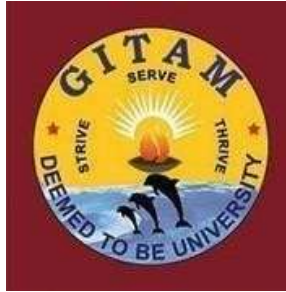
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CERTIFICATE

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ABSTRACT

This document describes the design of a School Management System, which aims to streamline the administrative and operational processes within a school. The system provides functionalities for teachers, students, and administrators to manage job applications, admissions, attendance, and announcements. It addresses the need for better organization and communication within the school community. Explore the abstract concept of a Learning Management System, a powerful tool that facilitates online learning, course management, and student engagement. The School Management System, presented in this document, is a comprehensive software solution meticulously designed to revamp and optimize the multifaceted administrative and operational processes inherent within educational institutions. This innovative system orchestrates a symphony of functionalities catering to teachers, students, and administrators, primarily focusing on the streamlining of job applications, admissions, attendance management, and timely announcements. It stands as an answer to the persistent call for heightened organization and seamless communication within the complex and dynamic ecosystem of a school community.

CHAPTER 1

INTRODUCTION

1. INTRODUCTION

In the ever-evolving landscape of education, the efficient management of school administrative processes is paramount to fostering a conducive learning environment. The traditional manual and paper-based methods no longer suffice in a world that thrives on digitalization and automation. To address this need, we present the School Management System, a web-based application designed to revolutionize how educational institutions handle their day-to-day operations.

The School Management System is an innovative solution aimed at streamlining administrative and operational tasks within schools. With a primary focus on improving communication, organization, and accessibility, this system caters to the needs of teachers, students, and administrators alike. By digitizing various aspects of school management, it offers a centralized hub for job applications, admissions, attendance tracking, and announcements. This digital transformation ensures greater efficiency, transparency, and accountability, ultimately enhancing the overall educational experience.

This document provides a comprehensive overview of the proposed system, detailing its key functionalities, user roles, and modules. It highlights the transition from the existing manual processes to a more robust, web-based platform, showcasing the potential benefits it can bring to educational institutions. While the system is not without its drawbacks, such as the need for enhanced security measures and authentication protocols, it holds the promise of bridging the gap between conventional practices and modern technological advancements.

As we delve deeper into the design and features of the School Management System, we will discover how it aims to empower teachers, students, and administrators to navigate the complexities of school management more efficiently, paving the way for an enhanced educational ecosystem.

CHAPTER 2

LITERATURE SURVEY

2. LITERATURE SURVEY

A literature survey, often referred to as a literature review, is a critical component of academic research and scholarly work. It involves an in-depth examination of existing literature, research, and publications related to a specific topic or research question. The purpose of a literature survey is to provide a comprehensive understanding of the current state of knowledge, identify gaps in research, and inform the direction and focus of a new study. In this literature survey, we will explore the key principles, methods, and significance of this essential research component.

Key Principles of Literature Survey:

Scope Definition: Before embarking on a literature survey, it's crucial to define the scope and boundaries of the study. Determine the specific research question or topic you aim to investigate, which will guide your search and analysis.

Information Sources: Literature surveys encompass a wide range of sources, including academic papers, books, articles, theses, and even gray literature. It's important to select relevant and credible sources for analysis.

Search Strategy: Establish a systematic search strategy to identify relevant literature. This may involve using academic databases, libraries, online journals, and search engines. Keyword selection and Boolean operators can help refine your search.

Critical Evaluation: Evaluate the quality and relevance of the sources you find. Consider the authority of the authors, the methodology used, and the publication's date.

Synthesis and Organization: Organize the literature into themes, concepts, or chronological order, depending on your research objectives. Synthesize the findings from each source to identify commonalities, contradictions, and gaps in the existing literature.

Citation and Referencing: Properly cite and reference the sources you consult. This is crucial for acknowledging the original authors and avoiding plagiarism.

Methods for Literature Survey:

Keyword Searching: Begin with a list of relevant keywords related to your research topic. Use these keywords to search academic databases, catalogs, and digital repositories.

Snowballing: Follow citations in papers and articles to discover additional sources that are related to your topic. This helps you uncover older or less-cited works.

Systematic Review: For more rigorous and structured literature surveys, consider following a systematic review methodology, which includes predefined criteria for source selection and quality assessment.

Mind Mapping: Create visual mind maps or diagrams to represent the relationships between different sources and ideas. This can help you organize the information and identify patterns.

Significance of Literature Survey:

Contextualizing Research: A literature survey places your research within the context of existing knowledge. It demonstrates that you are building upon previous work and contributing to the field.

Identifying Research Gaps: By analyzing existing literature, you can identify areas where research is lacking or where contradictory findings exist, paving the way for new research questions.

Avoiding Redundancy: A thorough literature survey ensures that your research is not duplicating existing work and that you are aware of what has already been explored.

Theoretical Framework: It provides the theoretical foundation for your research, helping you build a conceptual framework and theoretical underpinning.

Methodological Insights: Literature surveys can offer insights into the methodologies and research approaches used in your field, aiding in the selection of appropriate methods for your study.

CHAPTER 3

SYSTEM ANALYSIS

3. SYSTEM ANALYSIS

The Systems Development Life Cycle (SDLC), or Software Development Life Cycle in systems engineering, information systems and software engineering, is the process of creating or altering systems, and the models and methodologies that people use to develop these systems. In software engineering the SDLC concept underpins many kinds of software development methodologies.

Existing System

In the existing system, the school's administrative tasks are mostly manual and paper-based. Teachers have limited means of taking attendance, students lack easy access to their attendance records, and administrative tasks are time-consuming. There is also a lack of a centralized communication system for important notices.

Proposed system

The proposed School Management System is a web-based application that consists of three main user roles: Teacher, Student, and Admin. Here's a breakdown of the key modules and their functionalities:

SYSTEM MODULES:

MODULES DESCRIPTION:

Teacher:

Job Application: Teachers can apply for job positions, providing their details and qualifications.

Dashboard: After admin approval, teachers can access their dashboard.

Attendance Management: Teachers can take attendance for their classes and view attendance records later.

Notice Publication: Teachers can publish announcements and notices to students, such as assignment deadlines.

Student:

Admission/Signup: Students can apply for admission and create accounts.

Dashboard: After admin approval, students can access their dashboard.

Attendance: Students can view their own attendance records but cannot access other students' records.

Notice Viewing: Students can view announcements and notices posted by teachers or the admin.

Admin:

Account Creation: Admin can sign up for an account to access the system.

Application Approval: Admin can approve or reject teacher and student job/admission requests.

User Management: Admin can update teacher and student details, including password updates.

Notice Publication: Admin can publish announcements and notices for the entire school community.

Drawbacks:

Password Update: One drawback is that when updating teacher or student details, it is mandatory to update the password. This is a security measure to ensure data integrity.

Admin Role: Another drawback is that anyone can become an admin in the proposed system. In a real-world implementation, strict authentication and access control measures should be in place to prevent unauthorized individuals from becoming admins.

CHAPTER 4

IMPLEMENTATION

4. IMPLEMENTATION

The implementation phase of integrating Online Grocery technology with a farmer's portal for preserving the footage of selling and buying information of crops involves the following key steps:

- Infrastructure setup: Set up the necessary infrastructure to support the implementation of Online Grocery technology. This includes selecting a suitable Online Grocery platform or framework, setting up nodes, and configuring network connectivity.
- Smart contract development: Design and develop smart contracts that will govern the transactions and storage of selling and buying information. Define the data structures, functions, and business rules within the smart contracts to ensure accurate and secure recording of transactions.
- Integration with the farmer's portal: Integrate the developed smart contracts with the farmer's portal interface. Enable functionalities for farmers, vendors, and buyers to interact with the Online Grocery system, input transaction details, and access relevant information.
- Data management and encryption: Implement mechanisms for securely storing and encrypting the selling and buying information on the Online Grocery. Apply appropriate encryption techniques to protect sensitive data and ensure privacy.
- User interface development: Design and develop a user-friendly interface for the farmer's portal using the Python programming language. Create screens and functionalities for farmers to input selling information, buyers to submit buying information, and all users to view transaction history.
- Testing and quality assurance: Conduct thorough testing of the integrated system to ensure its functionality, reliability, and security. Perform unit testing, integration testing, and user acceptance testing to identify and address any bugs or issues.
- Deployment and user training: Deploy the integrated system to a production environment and provide necessary training to farmers, vendors, and buyers on how to use the farmer's

portal and interact with the Online Grocery system. Offer guidance on submitting transactions, accessing transaction records, and understanding the benefits of the Online Grocery technology.

- **Monitoring and maintenance:** Implement monitoring mechanisms to track the performance and security of the system in real-time. Continuously monitor the Online Grocery network, smart contracts, and user activities. Establish a maintenance plan to address any system updates, bug fixes, or security enhancements that may be required.
- **Documentation and knowledge transfer:** Document the implementation process, including system architecture, configurations, and user guides. Provide comprehensive documentation for future reference and knowledge transfer to ensure that stakeholders have the necessary information to maintain and operate the system effectively.
- **User support and feedback:** Offer ongoing user support and address any queries or issues raised by farmers, vendors, or buyers. Encourage feedback from users to identify areas for improvement and incorporate necessary enhancements to enhance user experience and system functionality.

By following these implementation steps, the integration of Online Grocery technology with the farmer's portal can be successfully executed, providing a secure and transparent platform for preserving the footage of selling and buying information of crops.

PYTHON

Python is a general-purpose interpreted, interactive, object-oriented, and high-level programming language. An interpreted language, Python has a design philosophy that emphasizes code readability (notably using whitespace indentation to delimit code blocks rather than curly brackets or keywords), and a syntax that allows programmers to express concepts in fewer lines of code than might be used in languages such as C++ or Java. It provides constructs that enable clear programming on both small and large scales. Python interpreters are available for many operating systems. CPython, the reference implementation of Python, is open source software and has a community-based development model, as do nearly all of its variant implementations. CPython is managed by the non-profit Python Software Foundation. Python features a dynamic type system and automatic memory management. It supports

multiple programming paradigms, including object-oriented, imperative, functional and procedural, and has a large and comprehensive standard library.

CHAPTER 5

SOFTWARE REQUIREMENT SPECIFICATION

5. SOFTWARE REQUIREMENT SPECIFICATION

5.1 Requirements Specification:

Requirement Specification provides a high secure storage to the web server efficiently. Software requirements deal with software and hardware resources that need to be installed on a server which provides optimal functioning for the application. These software and hardware requirements need to be installed before the packages are installed. These are the most common set of requirements defined by any operation system. These software and hardware requirements provide a compatible support to the operation system in developing an application.

5.1.1 HARDWARE REQUIREMENTS:

The hardware requirement specifies each interface of the software elements and the hardware elements of the system. These hardware requirements include configuration characteristics.

- System : Pentium IV 2.4 GHz.
- Hard Disk : 100 GB.
- Monitor : 15 VGA Color.
- Mouse : Logitech.
- RAM : 1 GB.

5.1.2 SOFTWARE REQUIREMENTS:

The software requirements specify the use of all required software products like data management system. The required software product specifies the numbers and version. Each interface specifies the purpose of the interfacing software as related to this software product.

Operating system	: Windows 10.
Coding Language	: Python.
Front-End	: Html, CSS.
Designing	: Html,css, javascript.
Data Base	: SQLite.

5.2 FUNCTIONAL REQUIREMENTS:

The functional requirement refers to the system needs in an exceedingly computer code engineering method.

The key goal of determinant “functional requirements” in an exceedingly product style and implementation is to capture the desired behavior of a software package in terms of practicality and also the technology implementation of the business processes.

5.3 NON FUNCTIONAL REQUIREMENTS

All the other requirements which do not form a part of the above specification are categorized as Non-Functional needs. A system perhaps needed to gift the user with a show of the quantity of records during info. If the quantity must be updated in real time, the system architects should make sure that the system is capable of change the displayed record count at intervals associate tolerably short interval of the quantity of records dynamic. Comfortable network information measure may additionally be a non-functional requirement of a system.

The following are the features:

- Accessibility
- Availability
- Backup
- Certification

- Compliance
- Configuration Management
- Documentation
- Disaster Recovery
- Efficiency(resource consumption for given load)
- Interoperability

5.4 PERFORMANCE REQUIREMENTS

Performance is measured in terms of the output provided by the application. Requirement specification plays an important part in the analysis of a system. Only when the requirement specifications are properly given, it is possible to design a system, which will fit into required environment. It rests largely with the users of the existing system to give the requirement specifications because they are the people who finally use the system. This is because the requirements have to be known during the initial stages so that the system can be designed according to those requirements. It is very difficult to change the system once it has been designed and on the other hand designing a system, which does not cater to the requirements of the user, is of no use.

The requirement specification for any system can be broadly stated as given below:

- The system should be able to interface with the existing system
- The system should be accurate
- The system should be better than the existing system

The existing system is completely dependent on the user to perform all the duties.

5.5 Feasibility Study:

Preliminary investigation examines project feasibility; the likelihood the system will be useful to the organization. The main objective of the feasibility study is to test the Technical, Operational and Economical feasibility for adding new modules and debugging old running system. All

systems are feasible if they are given unlimited resources and infinite time. There are aspects in the feasibility study portion of the preliminary investigation:

- Technical Feasibility
- Operation Feasibility

Economical Feasibility

5.5.1 Technical Feasibility

The technical issue usually raised during the feasibility stage of the investigation includes the following:

- Does the necessary technology exist to do what is suggested?
- Do the proposed equipments have the technical capacity to hold the data required to use the new system?
- Will the proposed system provide adequate response to inquiries, regardless of the number or location of users?
- Can the system be upgraded if developed?

Are there technical guarantees of accuracy, reliability, ease of access and data security?

5.5.2 Operational Feasibility

User-friendly

Customer will use the forms for their various transactions i.e. for adding new routes, viewing the routes details. Also the Customer wants the reports to view the various transactions based on the constraints. These forms and reports are generated as user-friendly to the Client.

Reliability

The package will pick-up current transactions on line. Regarding the old transactions, User will enter them in to the system.

Security

The web server and database server should be protected from hacking, virus etc

Portability

The application will be developed using standard open source software (Except Oracle) like Java, tomcat web server, Internet Explorer Browser etc these software will work both on Windows and Linux o/s. Hence portability problems will not arise.

Availability

This software will be available always.

Maintainability

The system uses the 2-tier architecture. The 1st tier is the GUI, which is said to be front-end and the 2nd tier is the database, which uses My-Sql, which is the back-end.

The front-end can be run on different systems (clients). The database will be running at the server. Users access these forms by using the user-ids and the passwords.

5.5.3 Economic Feasibility

The computerized system takes care of the present existing system's data flow and procedures completely and should generate all the reports of the manual system besides a host of other management reports.

It should be built as a web based application with separate web server and database server. This is required as the activities are spread throughout the organization customer wants a centralized database. Further some of the linked transactions take place in different locations.

CHAPTER 6

METHODOLOGY

6. Methodology

SDLC (Software Development Life Cycle) – Umbrella Model

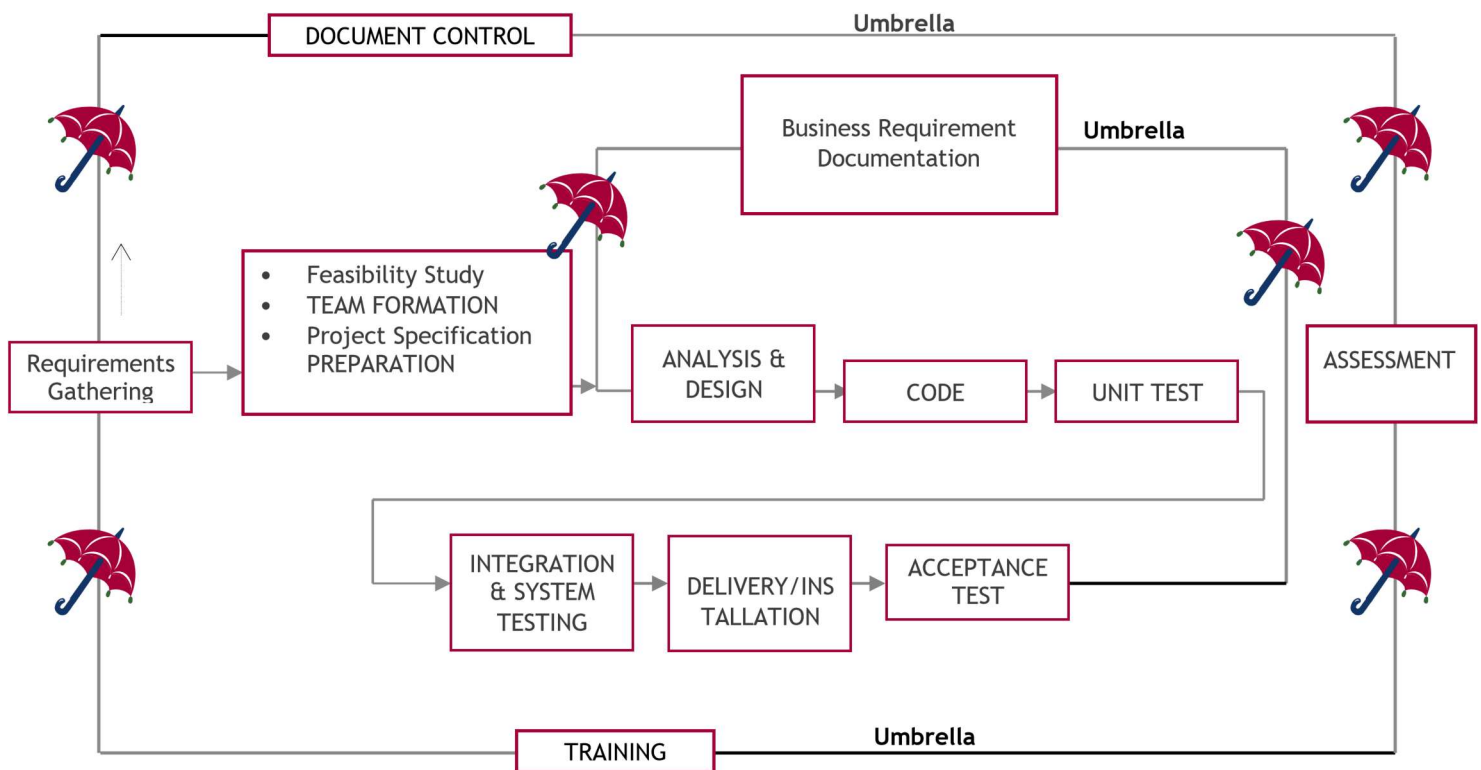


Fig no. 6.1 Umbrella model

SDLC is nothing but Software Development Life Cycle. It is a standard which is used by software industry to develop good software.

Requirements Gathering Stage

The requirements gathering process takes as its input the goals identified in the high-level requirements section of the project plan. Each goal will be refined into a set of one or more requirements. These requirements define the major functions of the intended application, define operational data areas and reference data areas, and define the initial data entities. Major functions include critical processes to be managed, as well as mission critical inputs, outputs and reports. A user class hierarchy is developed and associated with these major functions, data areas, and data entities. Each of these definitions is termed a Requirement. Requirements are identified by unique requirement identifiers and, at minimum, contain a requirement title and textual description.

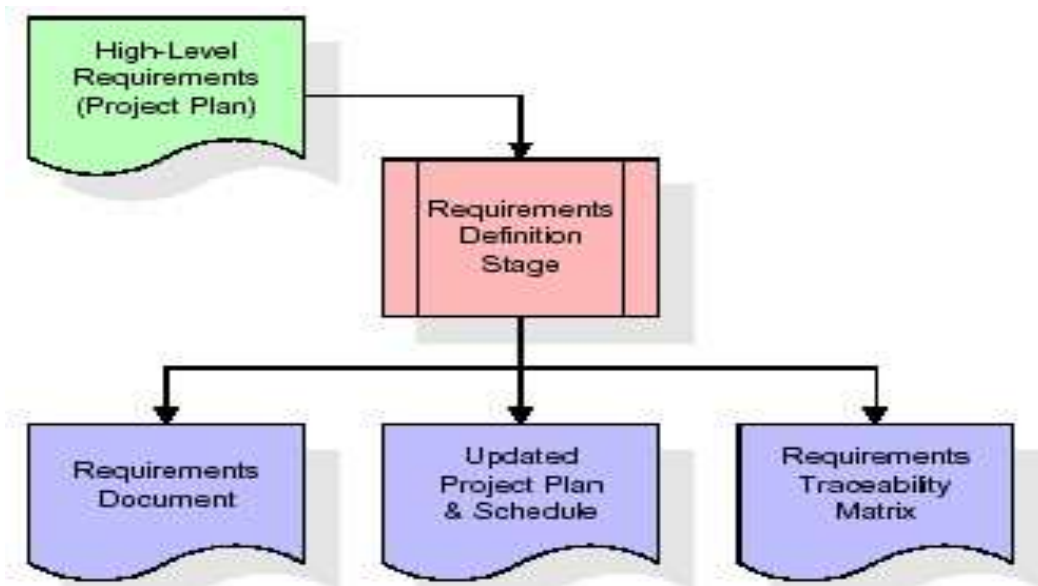


Fig no. 6.2 Requirements Gathering stage

These requirements are fully described in the primary deliverables for this stage: the Requirements Document and the Requirements Traceability Matrix (RTM). The requirements document contains complete descriptions of each requirement, including diagrams and references to external documents as necessary. Note that detailed listings of database tables and fields are not included in the requirements document.

The title of each requirement is also placed into the first version of the RTM, along with the title of each goal from the project plan. The purpose of the RTM is to show that the product

components developed during each stage of the software development lifecycle are formally connected to the components developed in prior stages.

In the requirements stage, the RTM consists of a list of high-level requirements, or goals, by title, with a listing of associated requirements for each goal, listed by requirement title. In this hierarchical listing, the RTM shows that each requirement developed during this stage is formally linked to a specific product goal. In this format, each requirement can be traced to a specific product goal, hence the term requirements traceability.

The outputs of the requirements definition stage include the requirements document, the RTM, and an updated project plan.

Feasibility study is all about identification of problems in a project, number of staff required to handle a project is represented as Team Formation, in this case only modules are individual tasks will be assigned to employees who are working for that project.

Project Specifications are all about representing of various possible inputs submitting to the server and corresponding outputs along with reports maintained by administrator.

Analysis Stage

The planning stage establishes a bird's eye view of the intended software product, and uses this to establish the basic project structure, evaluate feasibility and risks associated with the project, and describe appropriate management and technical approaches.

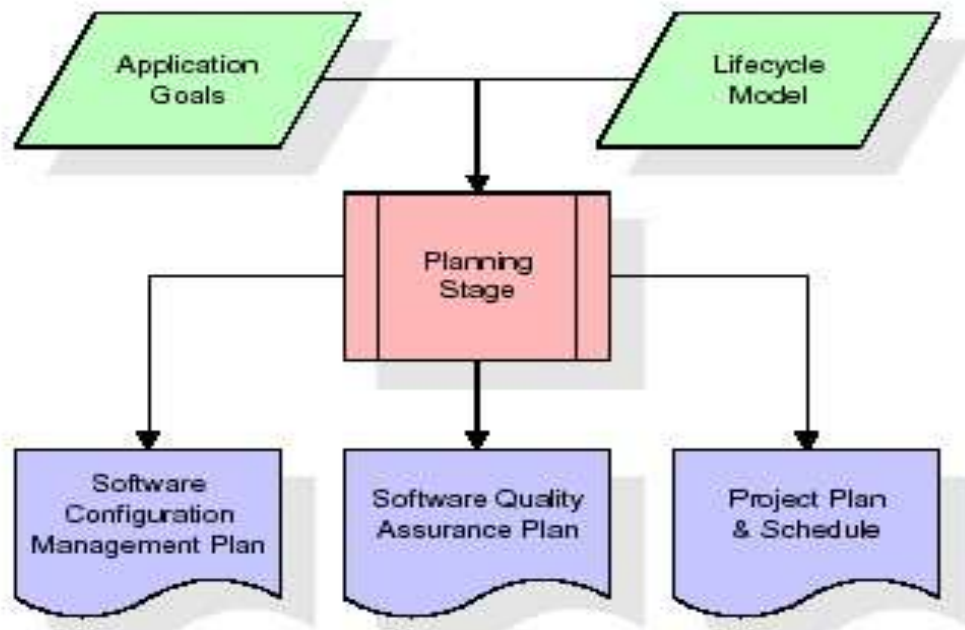


Fig no. 6.3 Analysis stage

The most critical section of the project plan is a listing of high-level product requirements, also referred to as goals. All of the software product requirements to be developed during the requirements definition stage flow from one or more of these goals. The minimum information for each goal consists of a title and textual description, although additional information and references to external documents may be included. The outputs of the project planning stage are the configuration management plan, the quality assurance plan, and the project plan and schedule, with a detailed listing of scheduled activities for the upcoming Requirements stage, and high level estimates of effort for the out stages.

Designing Stage

The design stage takes as its initial input the requirements identified in the approved requirements document. For each requirement, a set of one or more design elements will be produced as a result of interviews, workshops, and/or prototype efforts. Design elements describe the desired software features in detail, and generally include functional hierarchy diagrams, screen layout diagrams, tables of business rules, business process diagrams, pseudo code, and a complete entity-relationship diagram with a full data dictionary. These design elements are intended to describe

the software in sufficient detail that skilled programmers may develop the software with minimal additional input.

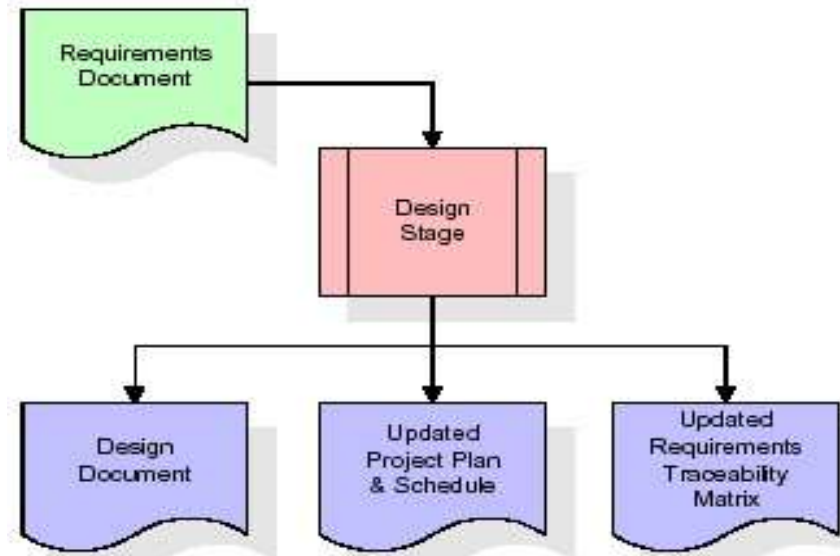


Fig no. 6.4 Designing stage

When the design document is finalized and accepted, the RTM is updated to show that each design element is formally associated with a specific requirement. The outputs of the design stage are the design document, an updated RTM, and an updated project plan.

Development (Coding) Stage

The development stage takes as its primary input the design elements described in the approved design document. For each design element, a set of one or more software artifacts will be produced. Software artifacts include but are not limited to menus, dialogs, data management forms, data reporting formats, and specialized procedures and functions. Appropriate test cases will be developed for each set of functionally related software artifacts, and an online help system will be developed to guide users in their interactions with the software.

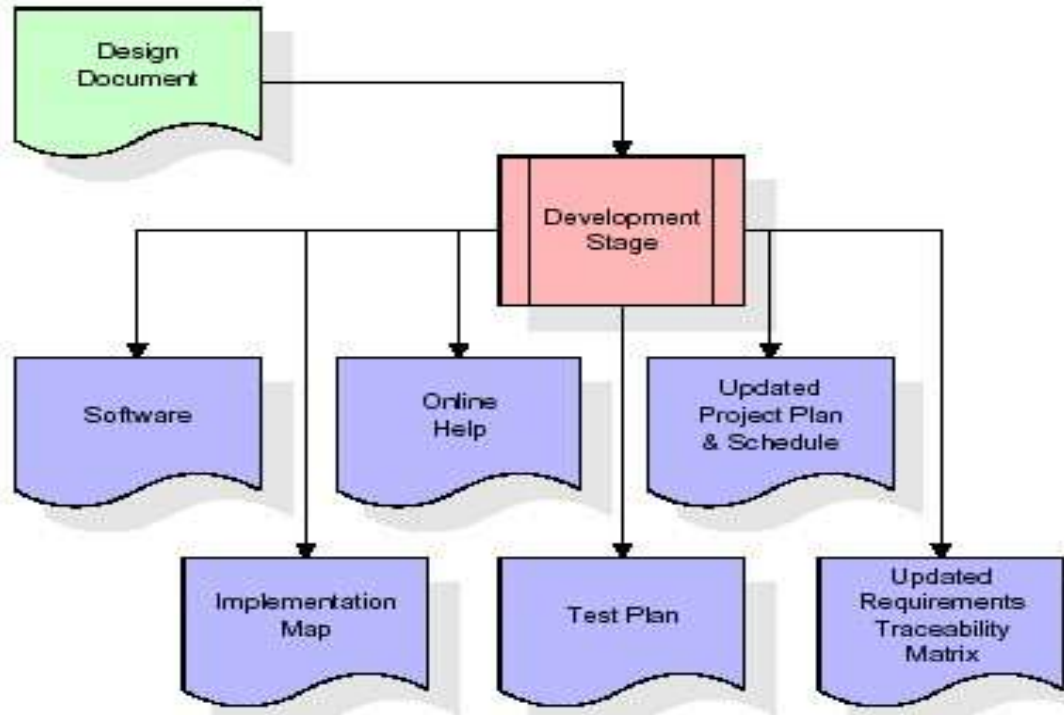


Fig no. 6.5 Coding stage

Integration & Test Stage

During the integration and test stage, the software artifacts, online help, and test data are migrated from the development environment to a separate test environment. At this point, all test cases are run to verify the correctness and completeness of the software. Successful execution of the test suite confirms a robust and complete migration capability. During this stage, reference data is finalized for production use and production users are identified and linked to their appropriate roles. The final reference data (or links to reference data source files) and production user list are compiled into the Production Initiation Plan.

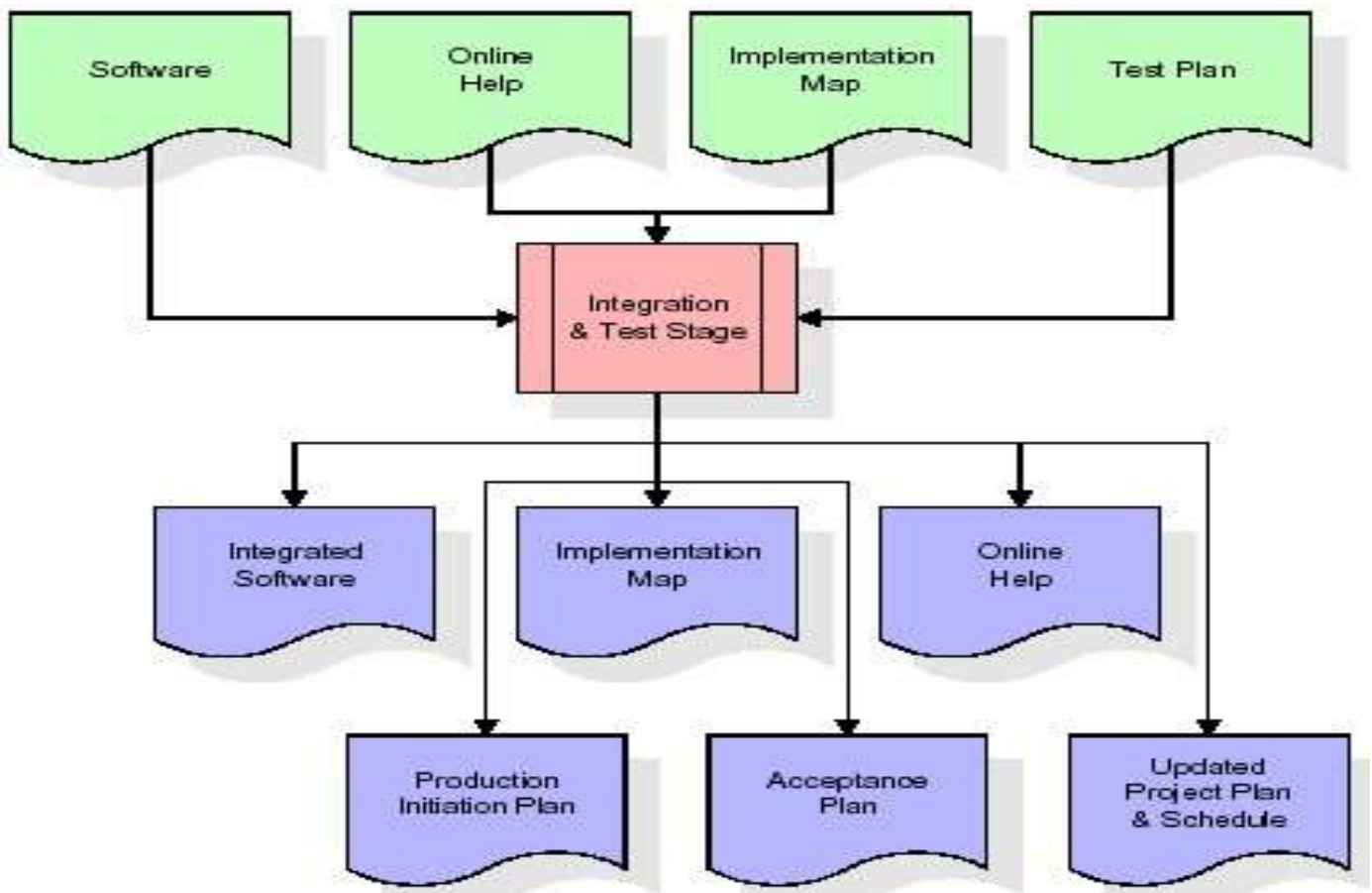


Fig no. 6.6 Integration and Testing Stage

Installation & Acceptance Test

During the installation and acceptance stage, the software artifacts, online help, and initial production data are loaded onto the production server. At this point, all test cases are run to verify the correctness and completeness of the software. Successful execution of the test suite is a prerequisite to acceptance of the software by the customer.

After customer personnel have verified that the initial production data load is correct and the test suite has been executed with satisfactory results, the customer formally accepts the delivery of the software.

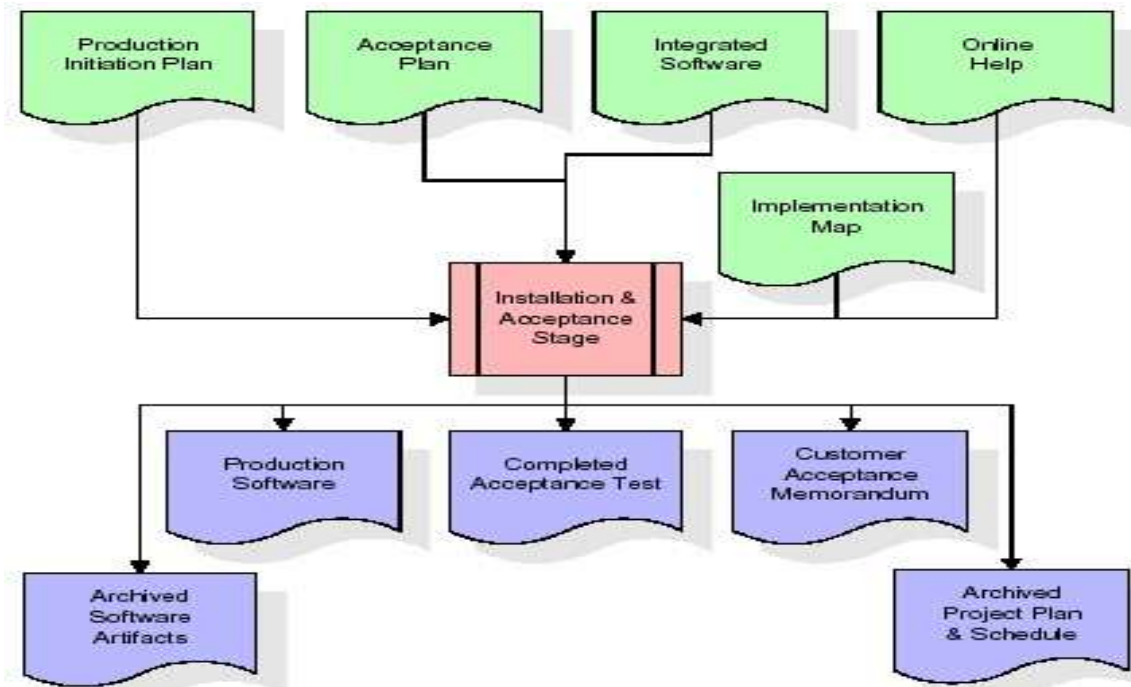


Fig no. 6.7 Installation

Maintenance

Outer rectangle represents maintenance of a project, Maintenance team will start with requirement study, understanding of documentation later employees will be assigned work and they will undergo training on that particular assigned category.

CHAPTER 7

SYSTEM DESIGN & UML DESIGN

7. System Design

7.1 DEFINITION

The most creative and challenging face of the system development is System Design. It provides the understanding and procedural details necessary for the logical and physical stages of development. In designing a new system, the system analyst must have a clear understanding of the objectives, which the design is aiming to fulfill. The first step is to determine how the output is to be produced and in what format. Second, input data and master files have to be designed to meet the requirements of the proposed output. The operational phases are handled through program construction and testing. Design of the system can be defined as a process of applying various techniques and principles for the purpose of defining a

device, a process or a system in sufficient detail to permit its physical realization. Thus, system design is a solution to “how to” approach to the creation of a new system. This important phase provides the understanding and the procedural details necessary for implementing the system recommended in the feasibility study. The design step provides a data design, architectural design, and a procedural design

7.2 OUTPUT DESIGN

A quality output is one, which meets the requirements of the end user and presents the information clearly. In any system results of processing are communicated to the users and to other system through outputs. In output design it is determined how the information is to be displaced for immediate need and also the hard copy output. It is the most important and direct source information to the user. Efficient and intelligent output design improves the system’s relationship to help user decision-making.

1. Designing computer output should proceed in an organized, well thought out manner; the right output must be developed while ensuring that each output element is designed so that people will find the system can use easily and effectively. When analysis design computer output, they should Identify the specific output that is needed to meet the requirements.

2. Select methods for presenting information.

3. Create document, report, or other formats that contain information produced by the system.

The output form of an information system should accomplish one or more of the following objectives.

- Convey information about past activities, current status or projections of the ❖ Future.
- Signal important events, opportunities, problems, or warnings.
- Trigger an action.
- Confirm an action.

7.3 INPUT DESIGN

The input design is the link between the information system and the user. It comprises the developing specification and procedures for data preparation and those steps are necessary to put transaction data in to a usable form for processing can be achieved by inspecting the computer to read data from written or printed document or it can occur by having people keying the data directly into the system. The design of input focuses on controlling the amount of input required, controlling the errors, avoiding delay, avoiding extra steps and keeping the process simple. The input is designed in such a way so that it provides security and ease of use with retaining the privacy. Input Design considered the following things:

- What data should be given as input?
- How the data should be arranged or coded?
- The dialog to guide the operating personnel in providing input.
- Methods for preparing input validations and steps to follow when error occur.

OBJECTIVES

1. Input Design is the process of converting a user-oriented description of the input into a computer-based system. This design is important to avoid errors in the data input process and show the correct direction to the management for getting correct information from the computerized system.
2. It is achieved by creating user-friendly screens for the data entry to handle large volume of data. The goal of designing input is to make data entry easier and to be free from errors. The data entry screen is designed in such a way that all the data manipulates can be performed. It also provides record viewing facilities.
3. When the data is entered it will check for its validity. Data can be entered with the help of screens. Appropriate messages are provided as when needed so that the user will not be in maize of instant. Thus the objective of input design is to create an input layout that is easy to follow

7.4 SYSTEM ARCHITECTURE

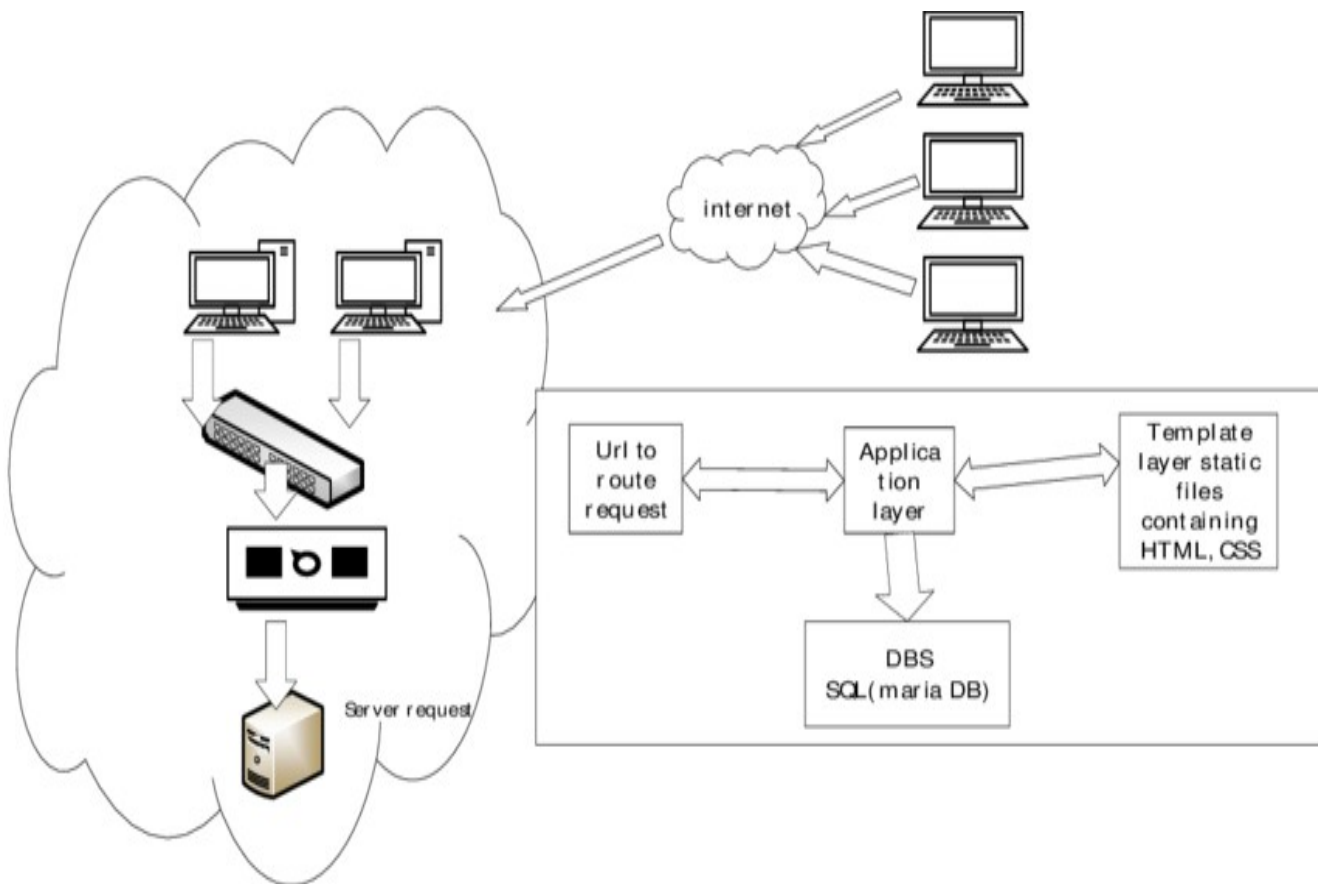


Fig: 1 System Architecture

7.5 DATA FLOW DIAGRAMS:

- The DFD is also called as bubble chart. It is a simple graphical formalism that can be used to represent a system in terms of input data to the system, various processing carried out on this data, and the output data is generated by this system.
- The data flow diagram (DFD) is one of the most important modeling tools. It is used to model the system components. These components are the system process, the data used by the process, an external entity that interacts with the system and the information flows in the system.

- DFD shows how the information moves through the system and how it is modified by a series of transformations. It is a graphical technique that depicts information flow and the transformations that are applied as data moves from input to output.
- DFD is also known as bubble chart. A DFD may be used to represent a system at any level of abstraction. DFD may be partitioned into levels that represent increasing information flow and functional detail.

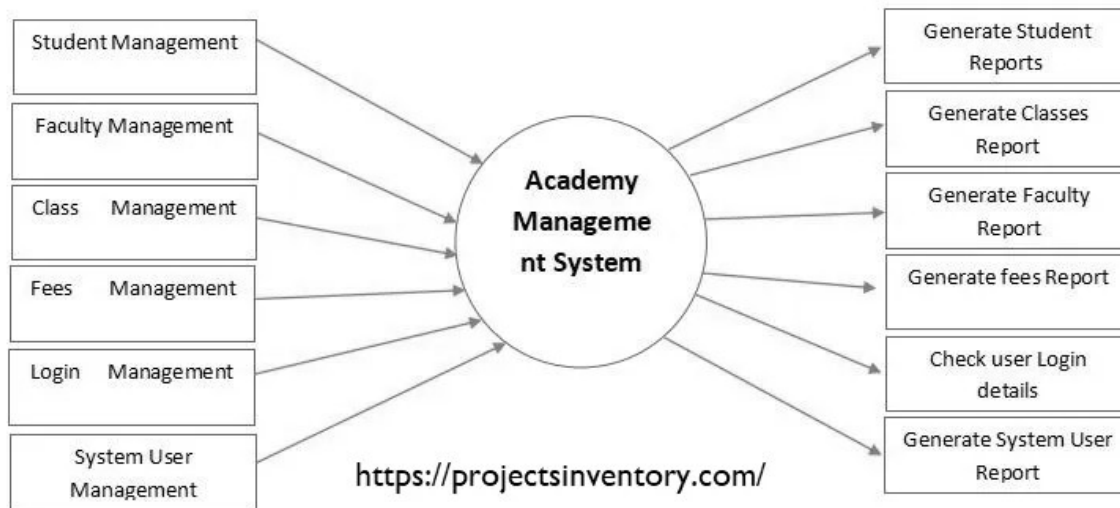


Fig: 2 Data Flow Diagram

7.6 UML DIAGRAMS

UML stands for Unified Modeling Language. UML is a standardized general-purpose modeling language in the field of object-oriented software engineering. The standard is managed, and was created by, the Object Management Group.

The goal is for UML to become a common language for creating models of object oriented computer software. In its current form UML is comprised of two major components: a Meta-model and a notation. In the future, some form of method or process may also be added to; or associated with, UML.

The Unified Modeling Language is a standard language for specifying, Visualization, Constructing and documenting the artifacts of software system, as well as for business modeling and other non-software systems.

The UML represents a collection of best engineering practices that have proven successful in the modeling of large and complex systems.

The UML is a very important part of developing objects oriented software and the software development process. The UML uses mostly graphical notations to express the design of software projects.

GOALS:

The Primary goals in the design of the UML are as follows:

- Provide users a ready-to-use, expressive visual modeling Language so that they can develop and exchange meaningful models.
- Provide extendibility and specialization mechanisms to extend the core concepts.
- Be independent of particular programming languages and development process.
- Provide a formal basis for understanding the modeling language.
- Encourage the growth of OO tools market.
- Support higher level development concepts such as collaborations, frameworks, patterns and components.
- Integrate best practices.

SEQUENCE DIAGRAM:

A sequence diagram in Unified Modeling Language (UML) is a kind of interaction diagram that shows how processes operate with one another and in what order. It is a construct of a Message Sequence Chart. Sequence diagrams are sometimes called event diagrams, event scenarios, and timing diagrams.

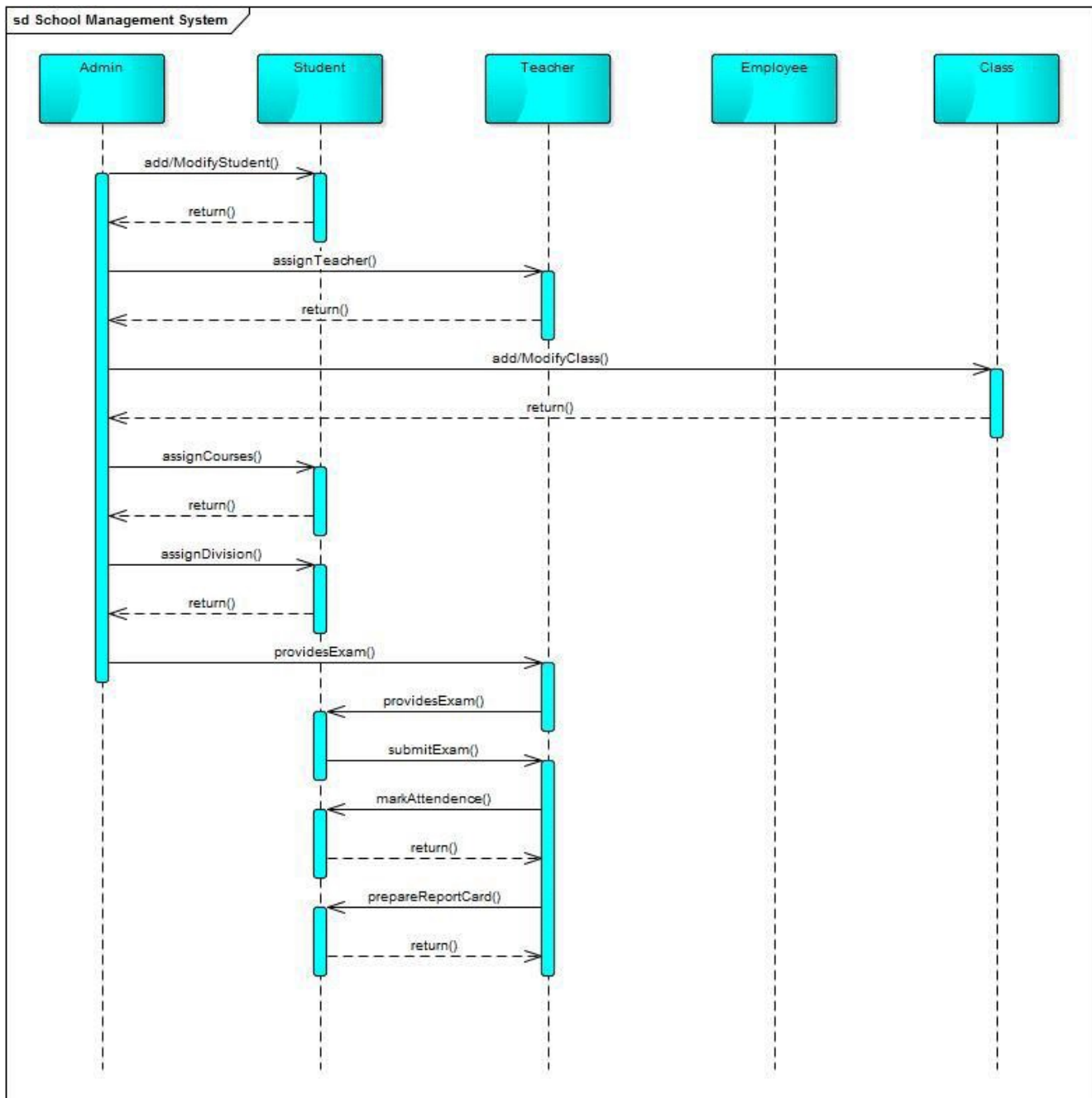


Fig: 3 Sequence Diagram

CLASS DIAGRAM:

In software engineering, a class diagram in the Unified Modeling Language (UML) is a type of static structure diagram that describes the structure of a system by showing the system's classes, their attributes,

operations (or methods), and the relationships among the classes. It explains which class contains information.

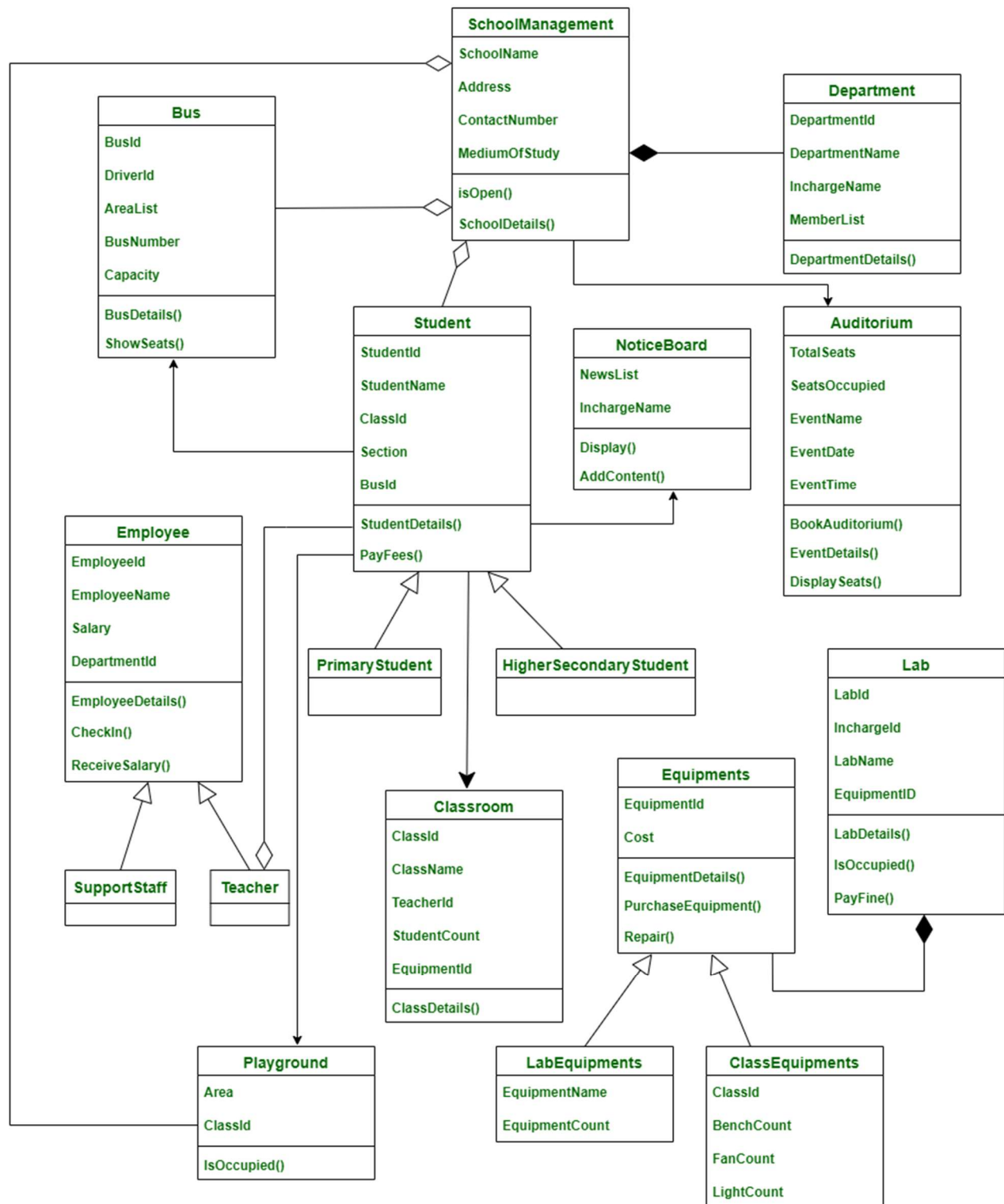


Fig: 4 Class Diagram

USE CASE DIAGRAM:

A use case diagram in the Unified Modeling Language (UML) is a type of behavioral diagram defined by and created from a Use-case analysis. Its purpose is to present a graphical overview of the functionality provided by a system in terms of actors, their goals (represented as use cases), and any dependencies between those use cases. The main purpose of a use case diagram is to show what system functions are performed for which actor. Roles of the actors in the system can be depicted.

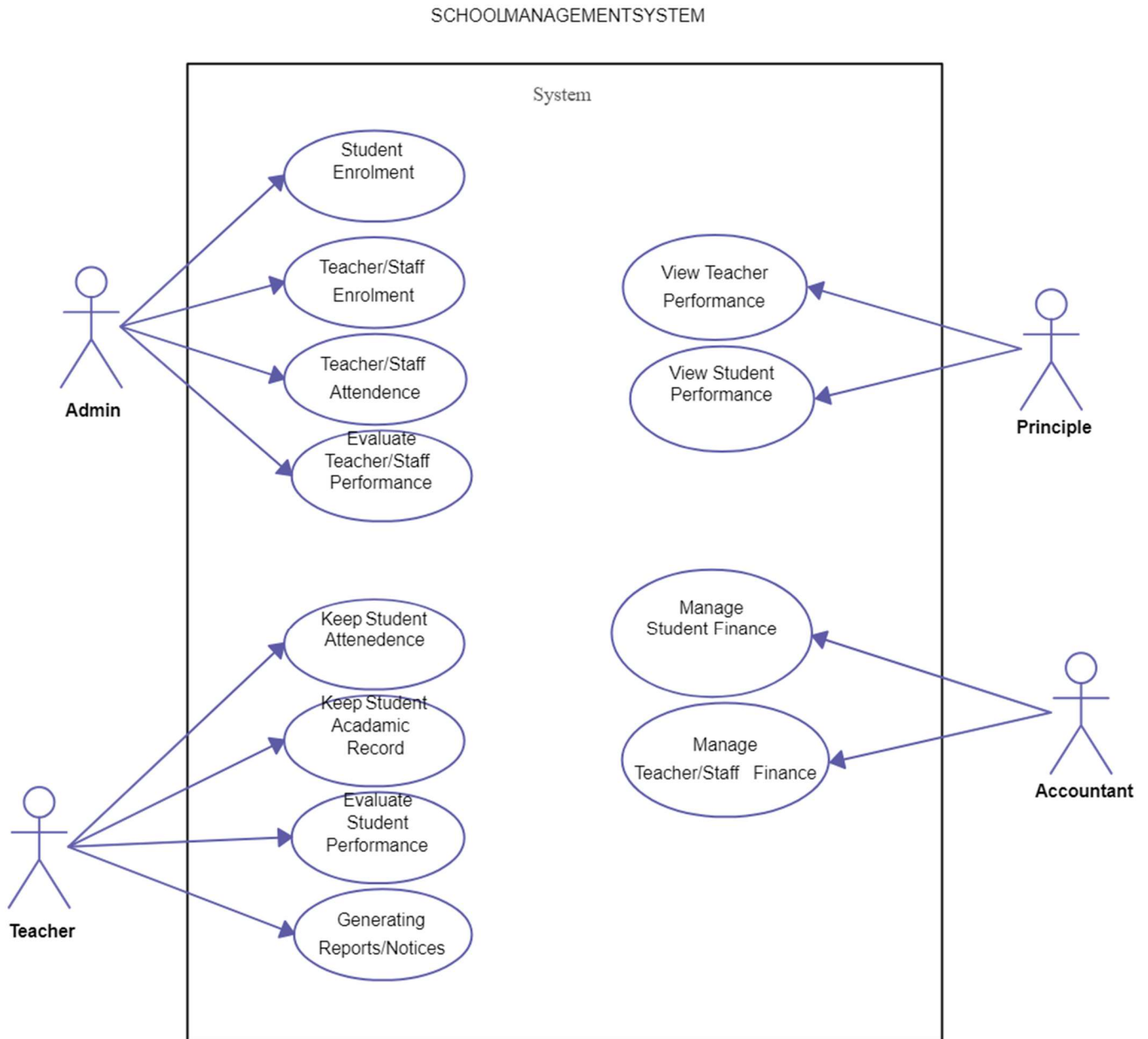


Fig: 5 Use-Case Diagram

ACTIVITY DIAGRAM

Activity diagrams are graphical representations of workflows of stepwise activities and actions with support for choice, iteration and concurrency. In the Unified Modeling Language, activity diagrams can be used to describe the business and operational step-by-step workflows of components in a system. An activity diagram shows the overall flow of control.

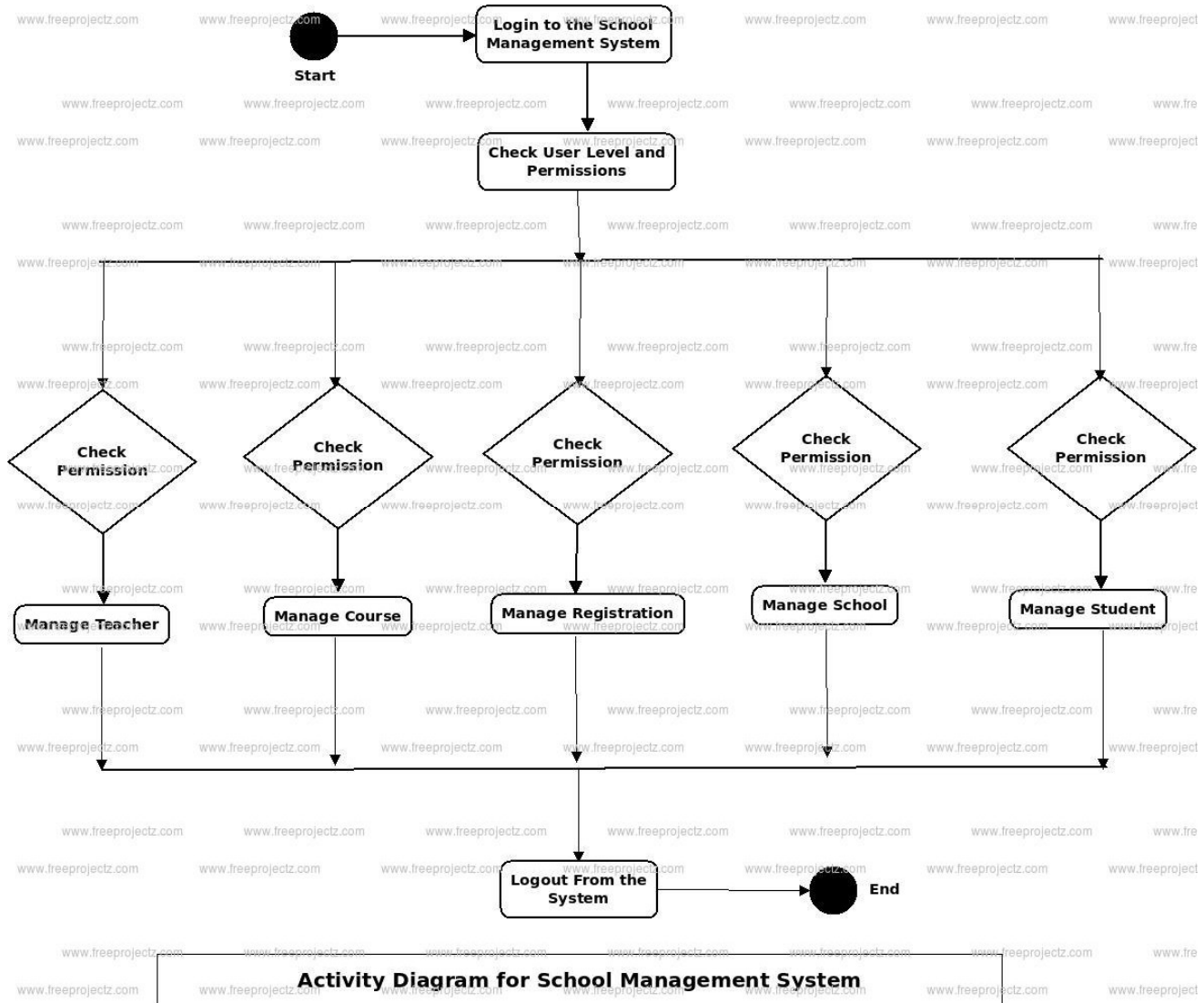


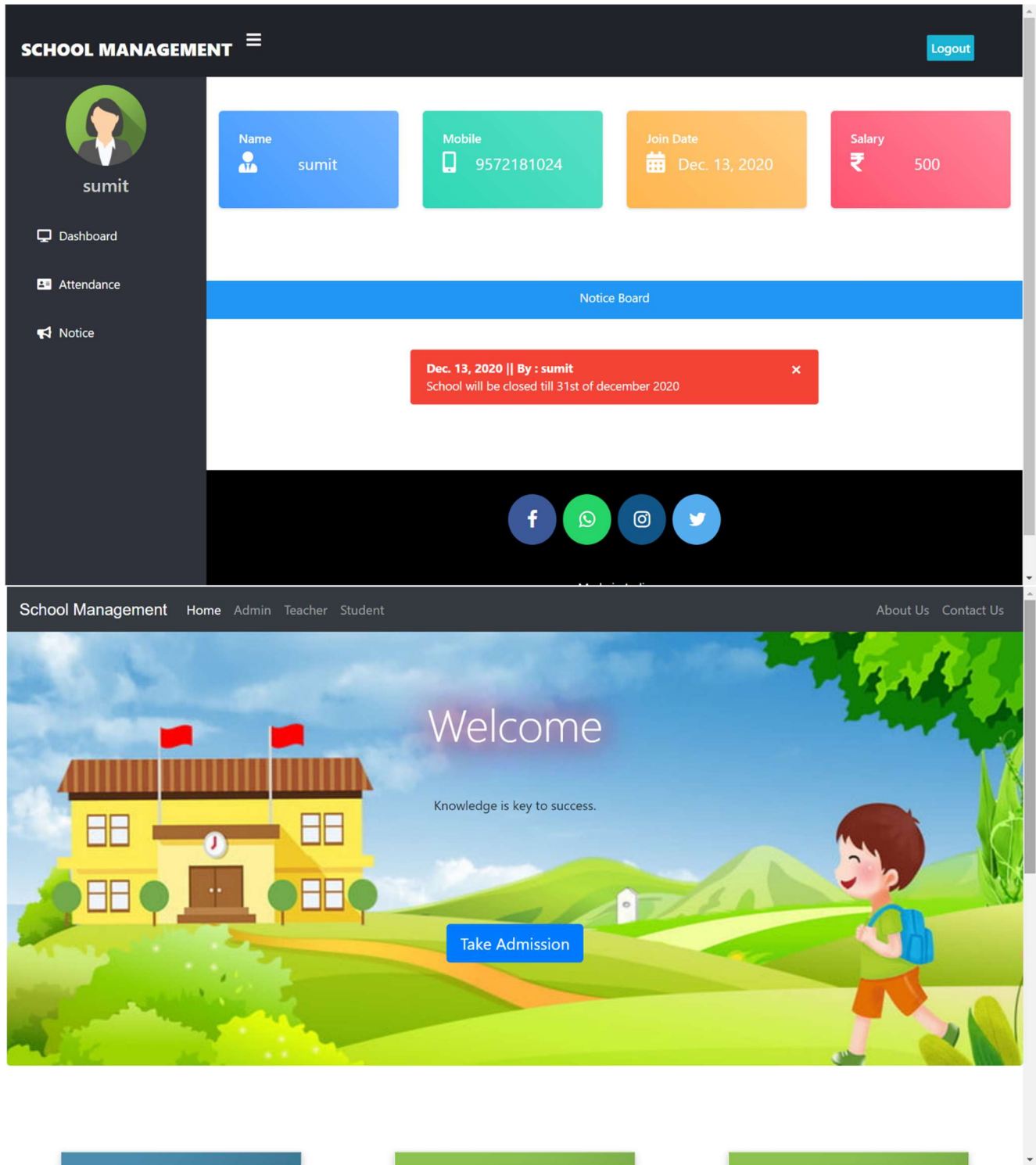
Fig: 6 Activity Diagram

7.7 Test Cases:

Test Case ID	Description	Test Data	Expected Result	Pass/Fail
TC001	Valid Login	Username: "user123", Password: "pass123"	Redirect to the user's dashboard.	Pass
TC002	Invalid Username	Username: "incorrectuser", Password: "pass123"	Display an error message.	Pass
TC003	Invalid Password	Username: "user123", Password: "wrongpass"	Display an error message.	Pass
TC004	Empty Username Field	Username: "", Password: "pass123"	Display an error message.	Pass
TC005	Empty Password Field	Username: "user123", Password: ""	Display an error message.	Pass
TC006	SQL Injection Attempt	Username: "user123' OR '1'='1", Password: "pass123"	Display an error message.	Pass
TC007	Excessive Password Length	Username: "user123", Password: "verylongpassword1234567890"	Display an error message.	Pass
TC008	Locked Account	Username: "lockeduser", Password: "pass123"	Display an account locked message.	Pass

CHAPTER 10

OUTPUT SCREENS



SCHOOL MANAGEMENT

Logout



sumit

Dashboard

Attendance

Notice

Attendance

Student Name	Present/Absent
shubham kumar	Present
prashant kumar	Absent
pintu kumar	Present
mohan kumar	Absent
sita kumari	Absent
12/12/2020	
submit	

SCHOOL MANAGEMENT

Logout



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Dashboard

Teacher

Student

Attendance

Fee

Notice

View All Student

Add Student

Approve Student

View Student Fee






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SCHOOL MANAGEMENT

Logout



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Dashboard

Teacher

Student

Attendance

Fee

Notice

Total Teacher1Pending Teacher0

Total Student1Pending Students0

Teachers Salary500Pending SalaryNone

Student Fee500Pending DuesNone

Notice Board

Dec. 13, 2020 ||By :sumit

School will be closed till 31st of december 2020

f

CHAPTER 11

CONCLUSION

11. CONCLUSION

The School Management System aims to enhance the efficiency and effectiveness of school operations by providing a centralized platform for managing job applications, admissions, attendance, and announcements. While there are some drawbacks to address, such as the admin role assignment, with proper security measures and user authentication, this system can significantly improve communication and organization within the school community.

Future Enhancements:

The School Management System holds great potential for further development and expansion to meet the evolving needs of educational institutions. While the system addresses key aspects of school management, there are several avenues for future enhancement and growth:

1. Enhanced Security Measures:

- Implement advanced security protocols and encryption techniques to safeguard sensitive data, including student and teacher information.

2. Multi-School Integration:

- Extend the system's capabilities to allow multiple schools within a district or region to use a centralized platform, facilitating data sharing and collaboration.

3. Parent and Guardian Portals:

- Introduce dedicated portals for parents and guardians, enabling them to monitor their child's attendance, academic progress, and receive important announcements.

4. Advanced Reporting and Analytics:

- Develop comprehensive reporting and analytics tools to provide administrators with valuable insights into school performance, resource allocation, and student outcomes.

5. Mobile Application:

- Create mobile apps for teachers, students, and administrators to ensure accessibility from anywhere, enhancing communication and productivity.

6. Online Payment and Fee Management:

- Incorporate secure online payment gateways for tuition fees, enabling parents to make payments electronically.

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